

PEAK ACHIEVERS
ONLINE ACADEMY

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MATHEMATICS

GRADE 8

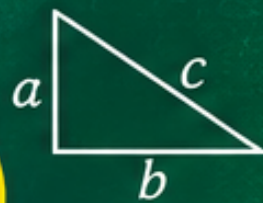
BOOK 2

- ✓ Revision Notes
- ✓ Worked Examples
- ✓ Application Questions
- ✓ Strand Quizzes
- ✓ Mixed Concepts Quizzes
- ✓ Five Final Tests (100 Marks Each)
- ✓ Complete Marking Schemes
- ✓ Real-Life and Competency-Based Questions
- ✓ Simple and Learner-Friendly Language



**CBE
ALIGNED**

$$a^2 + b^2 = c^2$$



$$C = 2\pi r$$

$$A = \pi r^2$$



$$a : b = \frac{a}{b}$$

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**MATHEMATICS EDITOR
& CONTENT VERIFIER**

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MATHEMATICS - GRADE 8

BOOK 2

ADVANCED REVISION & ASSESSMENT EDITION

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Mathematics Editorial Review

This manuscript was edited and academically verified by Kimani Peter, a qualified Mathematics teacher trained at Mount Kenya University, with review focused on mathematical accuracy, worked solutions, learner-level suitability and assessment consistency.

Dedication

In loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifice and inspiration remain part of my educational journey.

With heartfelt appreciation to Hyrine Kerubo, Peter Munde Mong'are, Teresa Kwamboka Sawe, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and all those whose support and encouragement have strengthened this journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart. ❤️

To my fellow teachers, guardians, family and every learner who continues to inspire the work behind Peak Achievers.

May every learner using this book remain curious, work with determination, and discover the confidence to learn, apply, think big and achieve.

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How to Use Book 2

This advanced revision and assessment edition uses brief question-and-answer revision, application quizzes, strand quizzes, mixed-concepts practice and five 100-mark final tests.

- Revise each key idea through the short Q&A notes.
- Attempt the Application Quiz before checking the answers.
- Complete the Strand Quiz after finishing each strand.
- Use the five Mixed Concepts Quizzes to connect ideas across strands.
- Attempt all five Final Tests under timed conditions, then use the marking schemes to check your work.

Contents

- Strand 1: Numbers - Integers; Fractions; Decimals; Squares and Square Roots; Rates, Ratio, Proportions and Percentages
- Strand 2: Algebra - Algebraic Expressions; Linear Equations
- Strand 3: Measurements - Circles; Area; Money
- Strand 4: Geometry - Geometrical Constructions; Coordinates and Graphs; Scale Drawing; Common Solids
- Strand 5: Data Handling and Probability - Data Presentation and Interpretation; Probability
- Mixed Concepts Quizzes 1-5
- Final Tests 1-5 - 100 marks each
- Answers and Marking Schemes

STRAND 1 - NUMBERS

1.1 Integers

Q: What are integers?

A: Whole numbers, their negatives and zero.

Q: How are integers ordered on a number line?

A: Values increase to the right and decrease to the left.

Q: What happens when a negative number is subtracted?

A: Subtracting a negative is equivalent to adding the corresponding positive number.

Q: What is the sign rule for multiplication and division?

A: Like signs give a positive answer; unlike signs give a negative answer.

Q: Where are integers used in daily life?

A: Temperature, lifts/floors, elevation, scores and financial changes.

Application Quiz

1. Arrange -8, -2, 0, 5, 7 in ascending order.
2. Evaluate $-12 + 7$.
3. Evaluate $18 - (-5)$.
4. Evaluate $(-6) \times 8$.
5. A temperature falls from 4°C to -3°C . Find the change.

Answers

1. -8, -2, 0, 5, 7
2. -5
3. 23
4. -48
5. A fall of 7°C .

1.2 Fractions

Q: What are equivalent fractions?

A: Fractions that represent the same value.

Q: How are unlike fractions added or subtracted?

A: Use a common denominator, then add or subtract the numerators.

Q: How are fractions multiplied?

A: Multiply numerators and denominators, then simplify.

Q: How is division by a fraction performed?

A: Multiply by the reciprocal of the divisor.

Q: How is a fraction of a quantity found?

A: Multiply the fraction by the quantity.

Application Quiz

1. Simplify $24/36$.
2. Calculate $3/5 + 7/10$.
3. Calculate $7/8 - 1/3$.
4. Calculate $4/9 \times 3/8$.
5. Calculate $5/6 \div 5/12$.
6. Find $2/5$ of KSh 1,500.

Answers

1. $2/3$
2. $1 \frac{3}{10}$
3. $13/24$
4. $1/6$
5. 2
6. KSh 600

1.3 Decimals

Q: What determines the value of a decimal digit?

A: Its place value: tenths, hundredths, thousandths and so on.

Q: How should decimal addition and subtraction be arranged?

A: Align decimal points vertically.

Q: How can a decimal be converted to a fraction?

A: Write it over the appropriate power of 10 and simplify.

Q: Why is estimation useful?

A: It helps detect unreasonable answers.

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MATHEMATICS - GRADE 8 - BOOK 2

FREE SAMPLE PACK

This free preview gives learners, parents and teachers a taste of the Peak Achievers revision approach: clear explanations, worked examples, application practice and competency-based assessment.

WHAT YOU GET IN THE FULL BOOK

- Book 2: Advanced Revision & Assessment Edition
- Brief Q&A; revision and application practice
- Mixed Concepts quizzes and national-exam-style final tests
- Full Book 2 price: KSh 400

SPECIAL BUNDLE: Book 1 + Book 2 = KSh 500

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